

LOAN DEFAULT PREDICTION

Final Capstone Report

From Predictive Modelling to Business-Ready Credit-Risk Decision Support

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Business objective: identify elevated-risk borrowers early enough to reduce credit losses while avoiding unnecessary restriction of satisfactory borrowers.

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LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation / Acronym	Meaning
API	Application Programming Interface
AUC	Area Under the Curve
EAD	Exposure at Default
EDA	Exploratory Data Analysis
F2-Score	F-Beta Score with Beta = 2
KNN	K-Nearest Neighbours
KYC	Know Your Customer
LGD	Loss Given Default
ML	Machine Learning
PR-AUC	Precision-Recall Area Under the Curve
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
ROI	Return on Investment

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1.0 Executive Summary

This capstone developed a business-ready credit-risk decision-support framework for identifying borrowers at elevated risk of loan default while preserving human underwriting judgement. The preferred solution is a tuned randomly oversampled Gradient Boosting model operating initially at a 0.34 enhanced-review threshold rather than as an automatic rejection rule.

On the independent test portfolio, the model identified 194 of 238 actual defaulters, equivalent to approximately 81.5% recall, while routing about 22.3% of applicants to enhanced review. It also identified approximately 81% of observed default-related loan exposure, indicating that the model can potentially help lenders focus review resources on a manageable portion of applications while capturing a substantial share of elevated credit risk.

The financial sensitivity analysis also suggested potential business value. Under the central illustrative assumptions of 50% Loss Given Default and 50% intervention effectiveness, potential avoided loss was approximately 878,600 monetary units. These figures remain scenario-based and should not be interpreted as guaranteed savings.

The principal recommendation is to progress the model to a controlled, bank-specific operational pilot with human credit oversight, risk-based review routing, fairness controls and continuous monitoring. Full deployment should occur only after the pilot confirms stable predictive performance, manageable review workload and positive realised business value.

1.1 Project Overview

The Loan Default Prediction capstone project was undertaken to develop a structured analytical approach for identifying borrowers who may be at elevated risk of loan default or severe delinquency. The project focused on supporting more informed and consistent credit-risk assessment by combining historical lending data, exploratory analysis and supervised machine-learning techniques.

The analysis was based on a historical portfolio of 5,960 previously approved loan applicants. The dataset contained borrower, credit-history, affordability, collateral and loan-related information, together with the historical repayment outcome used as the target for analysis. The project was developed in two broad stages. The first stage focused on Exploratory Data Analysis (EDA), data preparation and feature development. This stage examined the distribution of repayment outcomes and investigated how borrower and loan characteristics were associated with historical default. Particular attention was given to repayment capacity,

prior delinquency and derogatory credit behaviour, credit-history maturity, missing affordability information and collateral-related information.

The second stage focused on predictive modelling and validation. The data were divided into separate training, validation and independent test datasets so that model development decisions could be made without using the final test data. Five classification algorithms were evaluated: Logistic Regression, K-Nearest Neighbours, Decision Tree, Random Forest and Gradient Boosting. The project also examined several approaches to class imbalance, including the original class distribution, random oversampling, random undersampling and class weighting. The modelling process included feature engineering, leakage-safe preprocessing, model comparison, hyperparameter optimisation, operating-threshold analysis, model interpretation and one-time independent test evaluation. The intention was not simply to maximise predictive accuracy, but to identify a solution that could balance default detection, false-positive burden, probability quality, operational review capacity and business usability.

The final output of the project is therefore a credit-risk decision-support framework that translates model predictions into practical lending actions such as standard underwriting, enhanced verification, senior credit review and ongoing monitoring. The framework is designed to complement not replace existing underwriting policy, KYC, collateral assessment, affordability review and authorised human judgement.

1.2 Key Takeaways

A conventional classification threshold of 0.50 was not automatically adopted. Instead, operating-threshold analysis was conducted using validation data. A threshold of 0.34 was selected because it produced the strongest F2-score and balanced-accuracy trade-off within the evaluated range. At this threshold, the validation model identified 206 of 238 actual defaulters, while 62 satisfactory borrowers were sent for additional review. The model architecture, hyperparameters and threshold were locked before the independent test set was accessed.

On the final independent test portfolio, the model successfully identified 194 of 238 actual defaulters, equivalent to approximately 81.5% default recall. Only 44 actual defaults were missed, while 72 of 954 satisfactory borrowers were incorrectly flagged for elevated-risk intervention. The final model therefore retained strong discrimination, useful probability performance and a manageable intervention rate when applied to previously unseen borrowers.

Default recall	Enhanced review	Exposure proxy captured	Central avoided-loss scenario
81.5% 194 of 238	22.3% 266 of 1,192	≈81% 3.51m of 4.35m	878,600 monetary units

Source: Final capstone project analysis and notebook outputs.

The selected model also demonstrated meaningful potential business value. Of approximately 4,346,100 monetary units of approved-loan exposure associated with actual defaults, the model identified defaulters representing approximately 3,514,400 monetary units, or about 81% of observed default-related exposure. It routed 266 of 1,192 test applicants, approximately 22% of the test portfolio, to elevated-risk intervention.

Financial scenario analysis was used to avoid presenting unsupported savings claims. Under the central illustrative assumptions of 50% Loss Given Default and 50% intervention effectiveness, potential avoided loss was approximately 878,600 monetary units. Across the broader tested combinations of LGD and intervention effectiveness, illustrative potential avoided loss ranged from approximately 351,440 to 1,581,480 monetary units.

The financial analysis should not be interpreted as guaranteed savings. The available dataset does not contain institution-specific LGD, realised recovery rates, collection expenses, opportunity costs or intervention effectiveness. The results therefore provide a business decision framework, rather than a guaranteed financial return.

The broader business conclusion is that the value of the project extends beyond predictive accuracy. The model can potentially help a lender:

- identify elevated credit risk earlier;
- concentrate enhanced review on a manageable proportion of applications;
- reduce avoidable exposure to future defaults;
- improve the consistency of initial credit-risk assessment;
- strengthen the verification of incomplete affordability and collateral information;
- allocate experienced credit personnel to cases where their judgement is most valuable;
- and
- translate predicted probabilities into proportionate risk-management actions.

The solution should therefore be treated as a risk-prioritisation and decision-support system, not as an automatic approval or rejection mechanism.

1.3 Key Next Steps

The recommended next step is to progress the solution to a controlled operational pilot, rather than immediate full-scale deployment. The current 0.34 threshold should serve as the initial enhanced-review operating point during the pilot, while lower-risk applications continue through standard underwriting procedures. Elevated-risk cases should receive proportionate additional verification, manual review or senior credit assessment rather than automatic rejection.

Before production deployment, the adopting institution should recalibrate and independently validate the model using its own recent historical loan portfolio. Actual institutional information on Exposure at Default, LGD, recoveries, review costs, processing time, intervention effectiveness and opportunity costs should then replace the illustrative assumptions used in this capstone.

The pilot should also incorporate:

- graduated borrower-risk bands;
- enhanced verification of important affordability and collateral information;
- documented human credit judgement and override procedures;
- clear reason codes for elevated-risk assessments;
- fairness and bias monitoring;
- model-drift and calibration monitoring;
- measurement of false-positive and false-negative costs;
- actual review workload and processing-time measurement; and
- tracking of realised losses avoided and portfolio profitability.

The notebook specifically recommends ongoing monitoring of PR-AUC, ROC-AUC, recall, precision, false-positive and false-negative rates, probability calibration, risk-band distribution, realised default rates, intervention costs, actual losses avoided, overrides, borrower-group outcomes and changes in the applicant population.

Executive Decision

The project provides sufficient predictive, operational and scenario-based economic evidence to justify proceeding to a controlled bank-specific pilot. Full production deployment should occur only after the pilot confirms that the solution produces stable predictive performance, manageable review workload, acceptable borrower outcomes and positive realised business value under the institution's own operating conditions.

2.0 Problem and Solution Summary

2.1 Problem Being Solved

The project addresses the challenge of identifying borrowers who are at elevated risk of loan default or severe delinquency before losses materialise, while avoiding unnecessary adverse treatment of satisfactory borrowers.

Traditional credit underwriting can be slow, labour-intensive and inconsistent, particularly where applications require repeated manual review and where important affordability, collateral or credit-history information is incomplete. The business therefore needs a method that can support earlier and more consistent credit-risk assessment while preserving human judgement.

The exploratory analysis showed that historical default was associated with several borrower and credit characteristics, particularly previous delinquency and derogatory behaviour, debt-to-income burden, missing affordability information, credit-history maturity, cumulative adverse-credit history and selected missing collateral information. At the same time, apparently intuitive factors such as loan amount did not show a simple independent relationship with default, demonstrating the limitations of relying on individual underwriting rules or simple cut-offs.

The problem was therefore not simply to create a model with high predictive accuracy. The real business objective was to design a solution capable of:

- identifying a substantial proportion of future defaulters;
- limiting unnecessary intervention on satisfactory borrowers;
- producing meaningful and interpretable risk probabilities;
- supporting a manageable review workload;
- remaining stable on unseen applications; and
- providing a framework that can be translated into practical underwriting decisions and potential financial value.

The modelling objective consequently placed greater emphasis on default recall, precision, PR-AUC, specificity, probability calibration and business cost rather than accuracy alone. The earlier analytical findings specifically recognised that missing a genuine future defaulter can expose a lender to principal loss, collection costs and non-performing assets, while incorrectly flagging a satisfactory applicant may create lost business opportunities and poor customer experience.

2.2 Exploratory Data Analysis (EDA): What the Data Revealed

Before modelling, the historical portfolio was examined to understand the structure of default risk and to convert statistical patterns into business hypotheses. The EDA confirmed that the adverse-outcome class represented about one-fifth of the approved-loan portfolio and showed that repayment capacity, adverse credit behaviour, credit-history maturity and information completeness were more informative than simple loan-size rules.

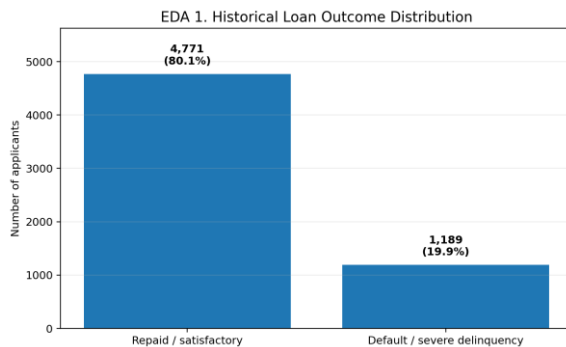


Figure 1. Historical outcome distribution

About 19.9% of the 5,960 approved applicants experienced default/severe delinquency. This class imbalance makes accuracy alone an unsuitable modelling objective.

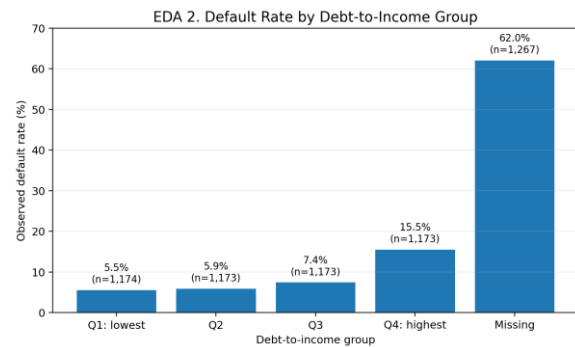


Figure 2. Debt-to-income and missing affordability information

Default increased in the highest debt-to-income group. Missing debt-to-income information was associated with an especially high historical default rate, supporting stronger affordability verification.

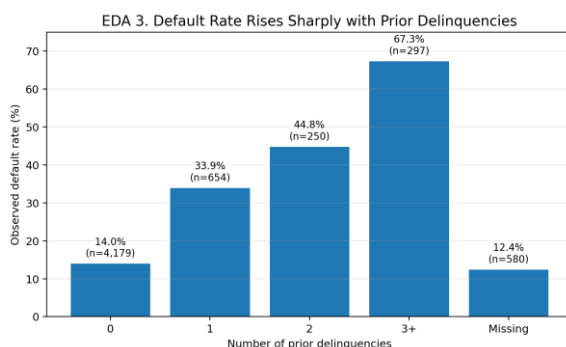


Figure 3. Prior delinquency and default

Observed default rose from about 14% among borrowers with no prior delinquency to about 67% among those with three or more, making adverse credit behaviour a clear warning signal.

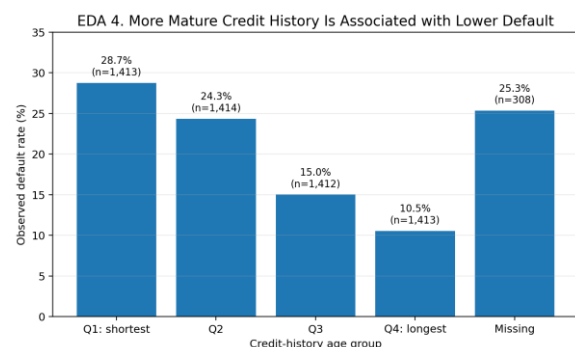


Figure 4. Credit-history maturity and default

Borrowers with shorter credit histories had materially higher default rates than those with longer histories, indicating that thinner or less mature credit records carry greater uncertainty.

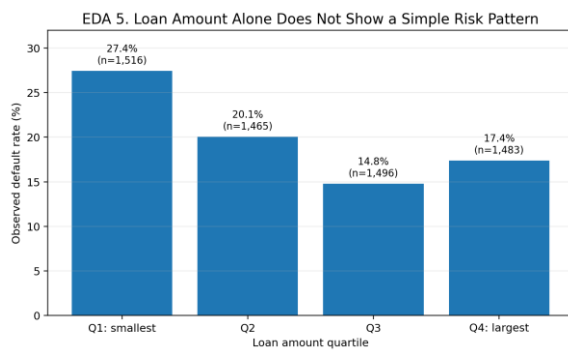


Figure 5. Loan amount quartiles

Loan size did not have a simple monotonic relationship with default. The smallest-loan quartile had the highest observed default rate, showing why multivariable assessment is preferable to a rule such as “larger loan = higher risk”.

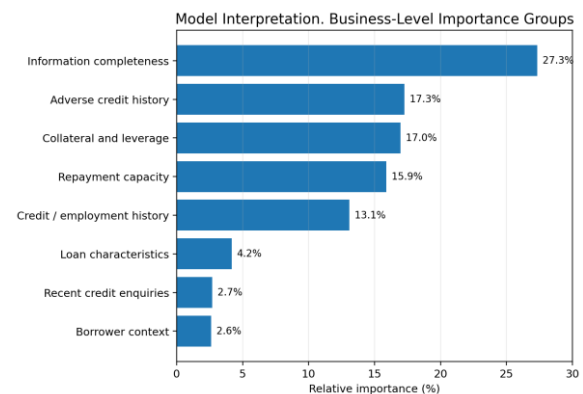


Figure 6. EDA themes carried into final model interpretation

The final model concentrated importance in information completeness, adverse credit history, collateral/leverage, repayment capacity and credit/employment history broadly consistent with the EDA hypotheses.

EDA conclusion. The exploratory evidence supported four business hypotheses for modelling: (1) repayment capacity matters; (2) previous adverse credit behaviour is a strong warning signal; (3) longer, more established credit history generally lowers uncertainty; and (4) missing or incomplete key financial information can itself identify cases needing verification. These findings were treated as predictive associations, not automatic rejection rules.

2.3 Final Proposed Solution Design

The final proposed solution is a human-supervised credit-risk decision-support framework built around a tuned randomly oversampled Gradient Boosting model. The solution is designed to operate as follows:

Loan application → data-quality and documentation checks → predictive default-risk score → risk classification → standard or enhanced review → human credit decision → outcome monitoring

The model itself does not make the final lending decision. Instead, it generates a probability of default that can be used to prioritise applications according to the level of predicted risk.

The selected model was developed through a rigorous sequence that included:

- Stratified train, validation and independent test separation;

- Leakage-safe preprocessing;
- Business-driven feature engineering;
- Comparison of five candidate classification algorithms;
- Explicit treatment of class imbalance;
- Hyperparameter optimisation;
- Validation-based operating-threshold analysis;
- Final model interpretation;
- One-time independent test evaluation; and
- Business and financial-value assessment.

The modelling framework deliberately maintained separate training, validation and test samples. Training data were used to estimate model parameters, validation data were used for model comparison and threshold selection, while the test set was reserved for final evaluation only. This reduces the risk of selecting a solution that performs well merely because development decisions were influenced by the final evaluation data.

Five candidate models were evaluated:

- Logistic Regression;
- K-Nearest Neighbours;
- Decision Tree;
- Random Forest; and
- Gradient Boosting.

Class-imbalance strategies included the original distribution, random oversampling, random undersampling and class weighting. Hyperparameter optimisation produced the strongest improvement in the oversampled Gradient Boosting model, increasing validation PR-AUC from approximately 0.84 to 0.90. The final selected configuration was therefore the tuned oversampled Gradient Boosting model.

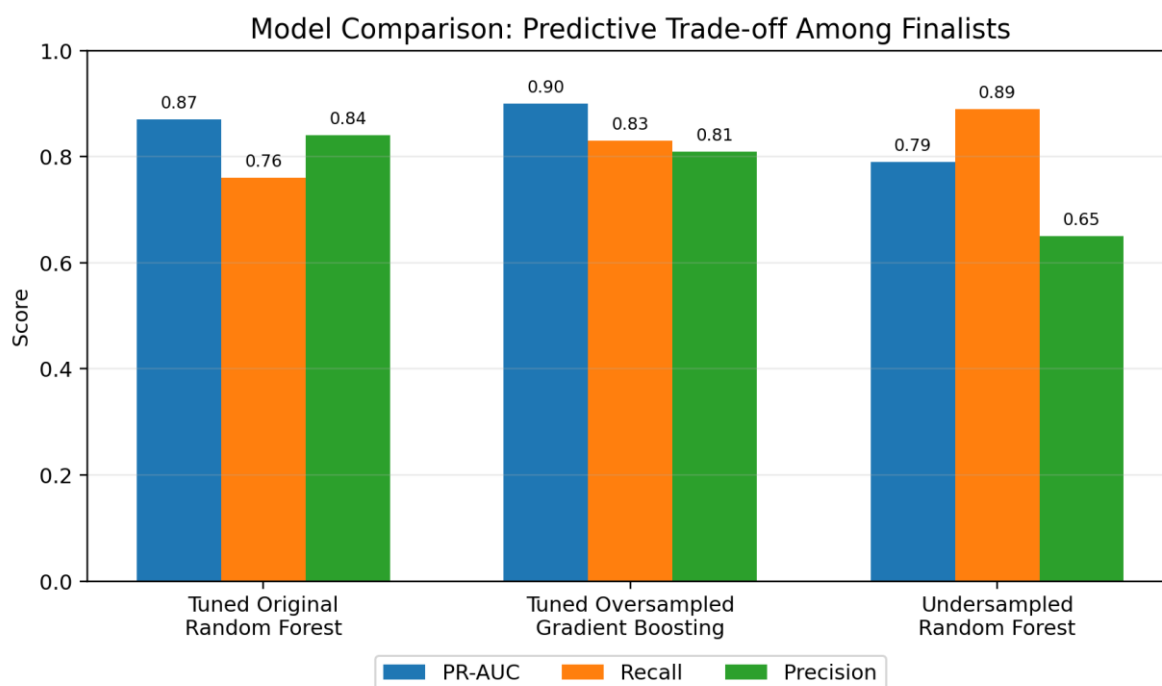


Figure 7. Finalist model comparison. The tuned oversampled Gradient Boosting model provided the strongest overall trade-off between ranking quality, default detection and false-alarm burden.

The selected model was not chosen because it had the highest recall alone. The undersampled Random Forest detected slightly more defaults, but at a much larger precision cost. The tuned oversampled Gradient Boosting model retained the strongest validation PR-AUC (about 0.90) while keeping recall, precision and probability quality at business-usable levels.

2.4 Operating Threshold and Risk Routing

A key part of the final solution design was the decision not to automatically use the conventional classification threshold of 0.50.

Validation analysis showed that a threshold of 0.34 provided a more appropriate business operating point. At this threshold, the model detected 206 of 238 validation defaults, while flagging 62 satisfactory borrowers for additional review. The model architecture, hyperparameters and threshold were then locked before the independent test set was accessed.

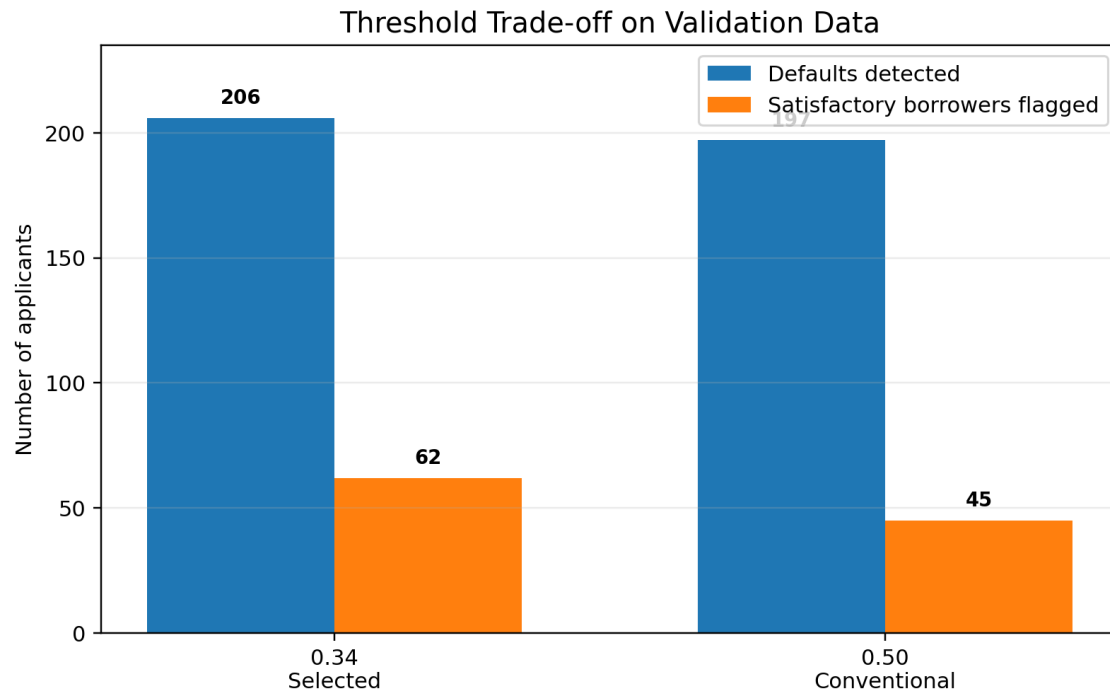


Figure 8. Threshold trade-off on validation data. Moving from 0.50 to 0.34 detected nine additional defaults while adding seventeen satisfactory-borrower reviews.

This threshold is therefore used as an enhanced-review trigger rather than a rejection rule. The extra review burden is deliberate: the business objective gives greater weight to avoiding missed defaults while still keeping the review queue manageable.

The 0.34 threshold is not proposed as an automatic rejection threshold.

Instead:

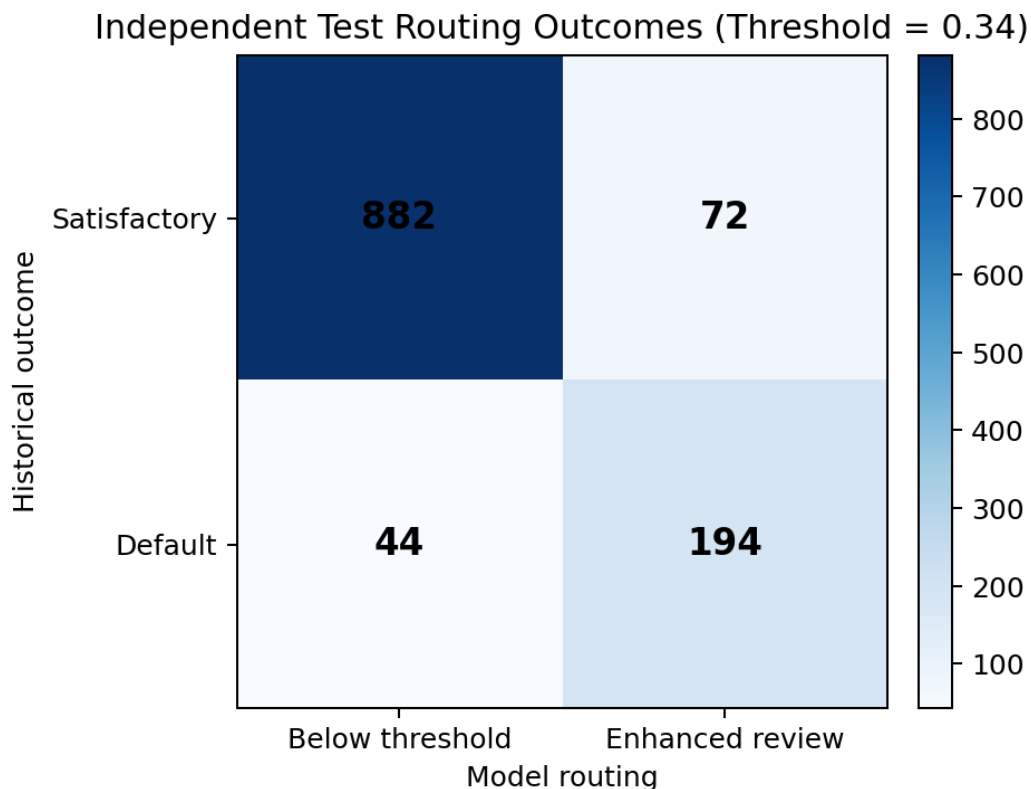
- applications below the threshold can continue through standard underwriting;
- applications at or above the threshold can be routed to enhanced verification or senior credit review;
- final lending decisions remain subject to existing credit policy, affordability assessment, verified information and authorised human judgement.

The notebook explicitly states that the model score should not independently determine approval or decline decisions, and that human judgement, verified borrower information and documented override procedures should remain part of the final process.

Risk probabilities can also be translated into graduated risk bands such as lower, moderate, high and very high risk, allowing the lender to apply different levels of review rather than relying on a single binary decision.

2.5 Why the Solution Is Valid

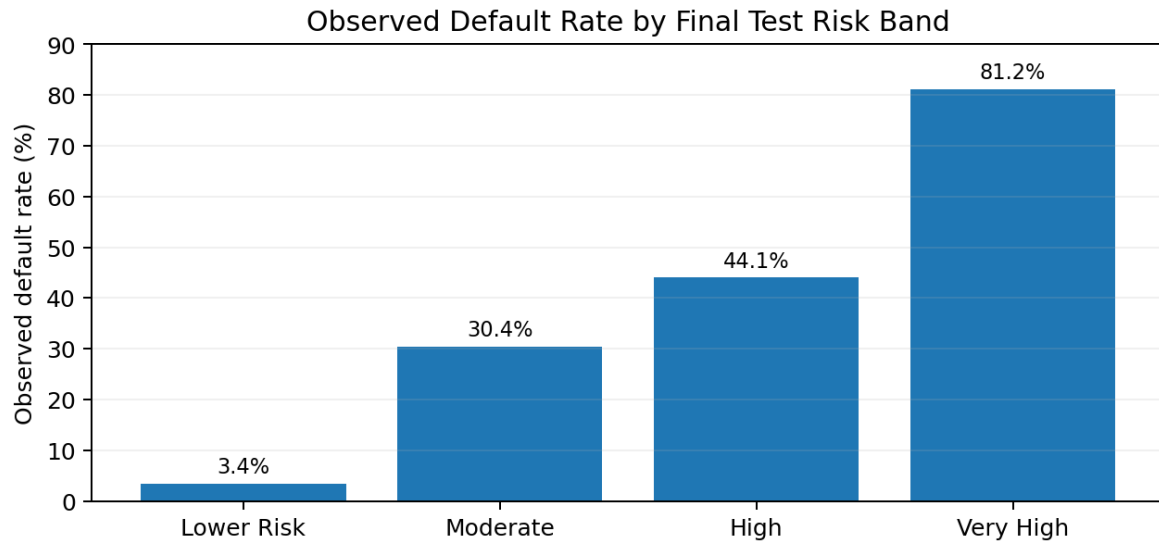
The solution is considered valid because its performance was demonstrated on an independent test dataset that was not used for model selection or threshold optimisation. On the final untouched test portfolio, the selected model identified 194 of 238 actual defaulters, achieved approximately 81.5% recall, missed 44 defaults and incorrectly flagged 72 of 954 satisfactory borrowers. This indicates that the model retained useful performance on previously unseen applications.



Source: Final capstone project analysis and notebook outputs.

Figure 9. Independent Test Loan-Risk Routing Outcomes Using the 0.34 Enhanced-Review Threshold

The model also produced meaningful risk separation. Observed default rates increased materially across the final risk bands, from 3.41% in the lower-risk group to 81.16% in the very-high-risk group. This supports the use of model probabilities for prioritising underwriting attention rather than relying only on a binary classification.



Source: Final capstone project analysis and notebook outputs.

Figure 10. Increasing Observed Default Risk Across Final Test Risk Bands

Model interpretation further strengthened the solution. The final model emphasised the same broad themes that emerged during exploratory analysis: information completeness, adverse credit history, collateral and leverage, repayment capacity, and credit/employment history. This consistency suggests that the model is learning plausible credit-risk relationships rather than depending solely on obscure statistical patterns.

2.6 Evidence of Business Relevance

On the independent test portfolio, the model identified defaulters representing approximately 3,514,400 of 4,346,100 monetary units of default-related approved-loan exposure, equivalent to roughly 81% of observed default-related exposure. It routed 266 of 1,192 applicants, approximately 22% of the test portfolio, to elevated-risk intervention.

This means the proposed framework can potentially allow the lender to concentrate more intensive credit-review resources on a relatively small portion of the portfolio while identifying a large proportion of historical defaults.

Illustrative financial analysis also suggests potential economic value. Under the central scenario of 50% LGD and 50% intervention effectiveness, estimated potential avoided loss was approximately 878,600 monetary units. However, the report treats this as a scenario rather than guaranteed savings, because institution-specific LGD, recovery, intervention cost and opportunity-cost information were not available.

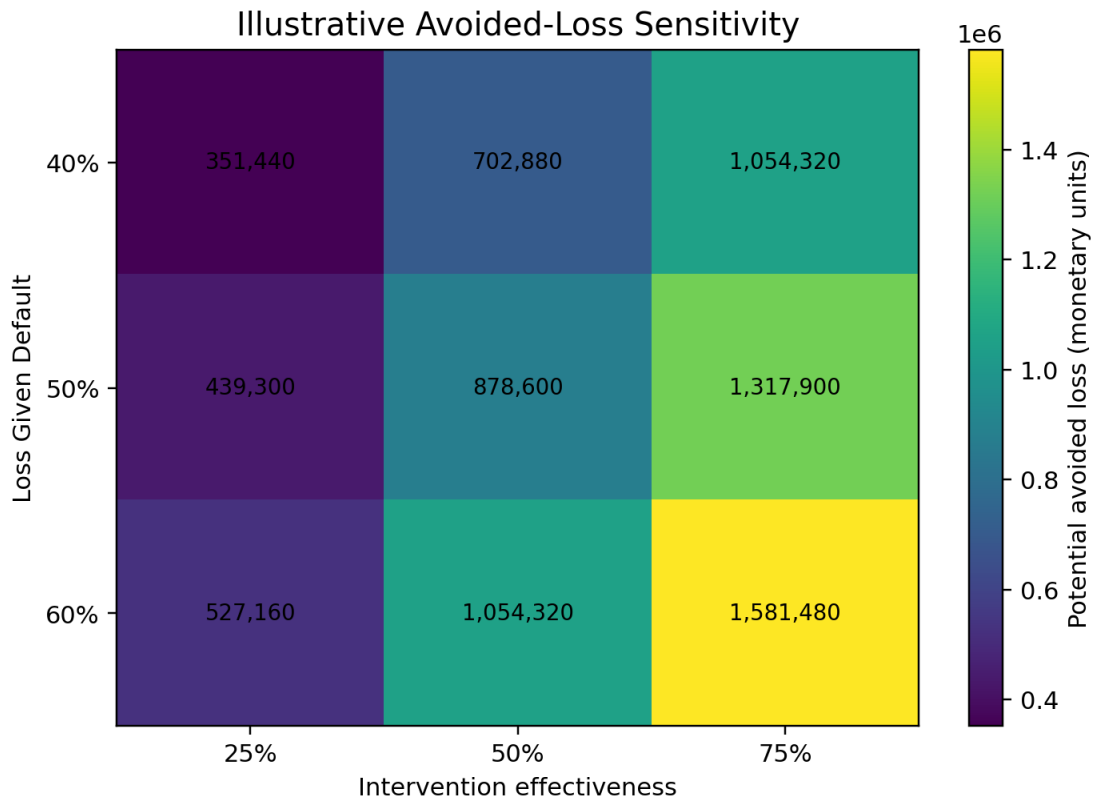


Figure 11. Illustrative avoided-loss sensitivity. Financial value depends jointly on Loss Given Default and how effective enhanced review/intervention is.

The central 50% LGD $\times$ 50% intervention-effectiveness scenario corresponds to potential avoided loss of approximately 878,600 monetary units. The sensitivity view is intentionally presented as a range because the dataset does not contain institution-specific LGD, recovery or intervention-effectiveness data.

2.7 Final Proposed Solution

A tuned oversampled Gradient Boosting credit-risk model operating initially at a 0.34 enhanced-review threshold, embedded within a human-supervised underwriting workflow and supported by risk bands, data verification, documented credit judgement, fairness controls and continuous monitoring.

The model is intended to help the lender determine:

- which applicants require enhanced verification;
- which cases should receive senior credit review;
- where additional collateral or documentation may be appropriate; and
- where closer post-disbursement monitoring may be justified.

The project therefore provides a valid and commercially relevant decision-support framework that is sufficiently strong to justify controlled operational piloting, subject to bank-specific recalibration, governance and ongoing monitoring.

3.0 Recommendations for Implementation

3.1 Recommended Implementation Strategy

The recommended implementation approach is a controlled operational pilot rather than immediate full-portfolio deployment. The pilot should use the tuned oversampled Gradient Boosting model with 0.34 as the initial enhanced-review trigger, while preserving existing underwriting policy and human credit authority.

S/N	Phase	Action	Decision gate
1.	Bank-specific validation	Recalibrate and validate on the institution's recent historical portfolio	Does predictive performance transfer?
2.	Limited operational pilot	Deployment in a defined product, branch set or business unit	Is review workload manageable?
3.	Business-value assessment	Measure processing time, costs, defaults detected and realized outcomes	Is there measurable value?
4.	Governance review	Assess fairness, overrides, drift, documentation and compliance	Is use responsible and controlled?
5.	Controlled scaling	Expand only after pilot success	Does the case support wider deployment?

The existing analysis supports this cautious approach. The final recommendation in the technical work is to proceed to a controlled pilot using the 0.34 operating threshold, risk bands, enhanced verification, documented human judgement and continuous monitoring.

3.2 Key Actionable Recommendations

Recommendation 1 - Use the model as decision support, not automatic rejection.

The model should not become an automatic approval or rejection engine.

It should be used to identify applicants requiring:

- Enhanced affordability verification;
- Additional collateral review;
- Senior underwriting attention;
- Documented exception handling; and
- Closer post-disbursement monitoring.

This approach is consistent with the model's intended role as a risk-prioritisation framework rather than a substitute for human judgement.

Recommendation 2 - Retain 0.34 as the initial pilot threshold

The 0.34 threshold should be used initially because validation analysis showed that it provided a stronger recall-weighted business trade-off than the conventional 0.50 threshold.

However, this threshold should remain subject to review once the bank has actual data on:

- Default losses;
- Staff review capacity;
- False-positive costs;
- Customer opportunity costs; and
- Intervention effectiveness.

The threshold should therefore be treated as an operational starting point, not a permanent fixed rule.

Recommendation 3 - Introduce Risk-Based Review Queues

Applicants should be routed into different underwriting paths based on predicted risk.

A practical design could include:

Lower risk — standard underwriting

Moderate risk — additional affordability checks

Elevated risk — enhanced credit review

High risk — senior credit assessment and documented decision

The earlier project recommendations already support the use of risk bands rather than a simple binary approve/reject classification.

Recommendation 4 - Improve Upfront Data Collection and Verification

The exploratory analysis identified important relationships involving missing affordability and collateral information. The bank should therefore strengthen the collection and verification of:

- Verified income;
- Debt obligations;
- Current property valuation;
- Mortgage information;
- Employment stability;
- Delinquency history; and
- Explanations for prior derogatory credit events.

This recommendation has value even independently of the predictive model because better input data can improve underwriting quality and reduce repeated requests for information during the application process.

Recommendation 5 - Measure Actual Business Value During the Pilot

The pilot should capture real institutional data for:

- Actual Exposure at Default;
- Realised Loss Given Default;
- Recovery rates;
- Collection costs;
- Review costs;
- Intervention effectiveness;
- Approval and rejection outcomes;
- Customer opportunity costs; and
- Post-intervention repayment performance.

This will allow management to replace the current scenario-based financial analysis with a bank-specific return-on-investment assessment.

Recommendation 6 - Establish Continuous Model Monitoring

The deployed solution should be monitored for:

- PR-AUC;

- ROC-AUC;
- Recall;
- Precision;
- False-positive rate;
- False-negative rate;
- Brier score and calibration;
- Realised default rate by risk band;
- Portfolio share by risk band;
- Model overrides;
- Intervention cost;
- Losses avoided;
- Borrower-group outcomes; and
- Changes in the applicant population.

3.3 Expected Benefits

The expected benefits of implementation can be grouped into five areas.

3.3.1 Credit-Risk Benefit

The final independent test model identified 194 of 238 actual defaulters, equivalent to approximately 81.5% default recall. This provides the lender with an opportunity to identify and respond to elevated risk before losses fully materialise.

3.3.2 Portfolio-Protection Benefit

The model identified approximately 3,514,400 of 4,346,100 monetary units of default-related approved-loan exposure, representing about 81% of observed default-related exposure. This demonstrates potential for more targeted risk mitigation.

3.3.3 Operational-Efficiency Benefit

Only 266 of 1,192 test applicants, approximately 22%, were routed to elevated-risk intervention. This suggests that specialist credit resources could be concentrated on the smaller group of applications where enhanced review is most valuable, rather than applying the same degree of intensive review to every case. Potential operational benefits include:

- Reduced manual review burden;
- Increased applications processed per credit officer;

- Earlier identification of additional documentation needs;
- More consistent initial risk assessment; and
- Better allocation of senior underwriting expertise.

3.3.4 Financial Benefit

Illustrative financial analysis produced potential avoided loss ranging from approximately 351,440 to 1,581,480 monetary units, depending on assumptions regarding LGD and intervention effectiveness. Under the central scenario of 50% LGD and 50% intervention effectiveness, potential avoided loss was approximately 878,600 monetary units.

The analysis also identified a break-even intervention cost of approximately 3,303 per flagged applicant under the central scenario.

These figures indicate potential economic value, but they remain illustrative rather than guaranteed.

3.3.5 Governance and Decision-Quality Benefit

The proposed framework provides a structured and documented method for identifying elevated-risk applicants.

This can improve:

- Decision consistency;
- Transparency;
- Auditability;
- Override control;
- Risk reporting; and
- Portfolio monitoring.

3.4 Expected Costs

Implementation will also create costs that must be considered.

Technology and Integration Costs

Potential costs include:

- Integration with the bank's loan-origination system;
- Data-pipeline development;
- Dashboard or API implementation;
- Server or cloud infrastructure;

- Cybersecurity controls; and
- Model-monitoring infrastructure.

Model Validation and Governance Costs

The bank will need resources for:

- Independent model validation;
- Periodic recalibration;
- Fairness testing;
- Internal audit;
- Model-risk management;
- Compliance review; and
- Documentation.

Staff and Change-Management Costs

Operational deployment may require:

- Training credit officers;
- Redesigning underwriting workflows;
- Developing review procedures;
- Defining escalation rules; and
- Monitoring staff overrides.

Enhanced-Review Costs

Because the model intentionally flags some satisfactory borrowers in order to capture more genuine defaults, the bank will incur additional review costs.

On the independent test portfolio, 72 satisfactory borrowers were incorrectly flagged for enhanced review. These costs must be balanced against the value of defaults detected.

3.5 Key Risks and Challenges

Risk / challenge	Potential consequence	Mitigation
Model drift	Performance may deteriorate as economic conditions, borrower behaviour or policy change	Monitor performance and recalibrate/retrain when triggers are met

False negatives	44 of 238 test defaults remained below the threshold	Retain conventional underwriting, monitoring and policy controls
False positives	Satisfactory borrowers may face extra review, delay or opportunity cost	Measure review burden; use proportionate rather than automatic adverse action
Data quality	Incomplete/outdated inputs can weaken predictions	Stronger upfront verification and data-quality controls
Fairness and bias	Contextual variables may create inappropriate proxy effects	Borrower-group monitoring, fairness tests, explainability and human review
Automation over-reliance	Staff may treat a score as a final answer	Document model limits and preserve human decision authority
Financial-estimation risk	Scenario value may differ from realised institutional value	Replace assumptions with actual EAD/LGD/recovery/cost data during pilot

3.6 Further Problems to Explore and Priority Order

The rubric explicitly asks what additional problems should be explored and in what priority order.

Priority	Further problem to explore	Why this comes next
1	Bank-specific external validation	Highest priority: confirm that the model transfers to the adopting bank's actual portfolio.
2	Actual EAD, LGD, recovery and intervention-cost modelling	Needed to convert prediction into reliable financial value.
3	Processing-time and workload analysis	Proves whether the system truly improves underwriting efficiency.

4	Fairness and bias assessment	Essential before wider production use in lending decisions.
5	Macroeconomic and model-drift analysis	Risk relationships may change across economic cycles.
6	Customer-impact analysis	Assesses delay, friction and opportunity cost for satisfactory applicants.
7	Champion-challenger model testing	Allows periodic comparison with alternative models.
8	Real-time integration / API deployment	Technology scaling should follow, not precede, evidence of predictive and business value.

The priority sequence is intentional: first establish whether the model works on the bank's actual portfolio; then establish its true economics; then optimise operations and technology.

3.7 Final Implementation Recommendation

The bank should proceed with a controlled, human-supervised operational pilot of the tuned oversampled Gradient Boosting solution.

The pilot should initially use the 0.34 enhanced-review threshold, with predicted risk translated into graduated underwriting actions rather than automatic rejection.

Implementation should include:

- Bank-specific validation;
- Improved upfront data collection;
- Risk-based review queues;
- Human credit authority;
- Documented overrides;
- Fairness controls;
- Continuous predictive monitoring;
- Actual financial-value measurement; and
- Periodic recalibration.

Full production deployment should occur only after the pilot demonstrates that the model provides:

1. Stable predictive performance;
2. Manageable review workload;
3. Acceptable fairness outcomes;
4. Measurable operational improvement; and
5. Positive realised financial value.

4.0 Conclusion

This capstone project demonstrates that loan-default prediction can be translated from a technical machine-learning exercise into a practical credit-risk decision-support framework for lending institutions.

The project began by analysing historical data from 5,960 previously approved loan applicants. Exploratory analysis showed that default risk was associated most strongly with factors such as repayment capacity, previous delinquency and derogatory credit behaviour, credit-history maturity, incomplete affordability information and aspects of collateral information. The findings also showed that loan amount alone did not provide a simple or reliable indication of default risk, reinforcing the need for a multivariable predictive approach.

A rigorous modelling process was then applied using separate training, validation and independent test datasets. Five classification algorithms were compared, alternative class-imbalance treatments were evaluated, and hyperparameter optimisation and operating-threshold analysis were conducted using development data only. The final model architecture and operating threshold were locked before evaluation on the independent test dataset, thereby providing a credible assessment of how the solution performs on previously unseen borrowers. The final selected model was a tuned randomly oversampled Gradient Boosting classifier, operated initially at a probability threshold of 0.34. On the independent test portfolio, the model correctly identified 194 of 238 actual historical defaulters, representing approximately 81.5% recall, while missing 44 defaults and incorrectly flagging 72 of 954 satisfactory borrowers.

From a business perspective, the model routed approximately 22% of the test portfolio to elevated-risk intervention while identifying about 81% of the approved-loan exposure associated with actual defaults. This suggests that the solution could help lenders concentrate enhanced credit-review resources on a relatively small proportion of applications while still detecting a substantial share of elevated-risk borrowers.

The financial analysis also indicates potential value. Under the central illustrative scenario of 50% Loss Given Default and 50% intervention effectiveness, potential avoided loss was

approximately 878,600 monetary units. However, these estimates are scenario-based and should not be interpreted as guaranteed financial savings because institution-specific recovery, LGD, intervention-cost and opportunity-cost information was unavailable.

The principal conclusion of the project is therefore that predictive performance alone is not sufficient for successful lending implementation. A useful credit-risk solution must combine predictive capability with:

- Manageable review workload;
- Understandable risk drivers;
- Suitable operating thresholds;
- Human credit judgement;
- Reliable borrower information;
- Fairness and governance controls;
- Continuous monitoring; and
- Measurable business value.

The model should therefore not be used as an automatic loan approval or rejection mechanism. Instead, it should support the prioritisation of credit-review effort, additional verification and senior underwriting attention.

Accordingly, the final recommendation is to proceed to a controlled bank-specific operational pilot. The pilot should validate the model on the institution's own recent portfolio, maintain the 0.34 threshold initially as an enhanced-review trigger, establish graduated risk-based review processes, retain documented human decision authority, and measure both predictive and financial outcomes. Full-scale deployment should occur only after the institution confirms stable model performance, acceptable fairness outcomes, manageable operational costs and positive realised business value.

Appendix A - Selected Model Performance

Summary

Metric	Validation (0.34)	Independent test (0.34)	Business interpretation
PR-AUC	≈0.90	≈0.86	Strong ranking performance for the minority/default class
ROC-AUC	≈0.96	≈0.95	Strong discrimination between outcomes
Recall	≈0.87	≈0.82	Captures most actual defaults
Precision	≈0.77	≈0.73	Most flagged cases are genuinely elevated risk, but review cost remains
Specificity	≈0.94	≈0.92	Most satisfactory borrowers remain below the threshold
F2-score	≈0.84	≈0.80	Supports the recall-weighted business objective
Balanced accuracy	≈0.90	≈0.87	Good performance across both outcome classes
Brier Score	≈0.06	≈0.07	Useful probability quality
Defaults detected	206 of 238	194 of 238	Strong default capture
Good borrowers flagged	62	72	Operational review burden to manage
Portfolio flagged	22.48%	22.32%	Enhanced review concentrated on roughly one-fifth of applicants

Source: Final capstone project analysis and notebook outputs.

A.1 Final Test Risk Bands

Risk band	Definition	Applicants	Observed defaults	Observed default rate
Lower Risk	< 0.20	880	30	3.41%
Moderate	$0.20 - < 0.34$	46	14	30.43%
High	$0.34 - < 0.60$	59	26	44.07%
Very High	≥ 0.60	207	168	81.16%

Source: Final capstone project analysis and notebook outputs.

Appendix B - Proposed Operational Workflow

The model should be embedded inside an existing credit process rather than used as a stand-alone decision engine. The proposed pilot workflow is:

1. Application submitted

Borrower submits application and required supporting documents.

2. Data-quality checks

Verify key affordability, collateral, employment and credit-history fields; resolve critical missing data where possible.

3. Model scoring

Generate probability of default using the validated bank-specific pipeline.

4. Risk-band assignment

Translate the probability into lower, moderate, high or very-high risk.

5. Review routing

Below 0.34: standard underwriting. At/above 0.34: enhanced verification and/or senior review.

6. Human credit decision

Authorised credit personnel make the final decision using model output alongside policy, KYC, collateral, affordability and professional judgement.

7. Outcome monitoring

Track repayment, default, override, review time, cost and portfolio outcomes.

8. Governance feedback

Use actual outcomes to monitor drift, fairness, calibration and the need for recalibration or threshold review.

Operating principle

The model prioritises human attention; it does not replace KYC, collateral verification, legal/compliance review or authorised credit judgement.

Appendix C - Key Limitations and Monitoring Indicators

C.1 Key Limitations

- The dataset represents historical previously approved applicants and may not fully represent future applicants or rejected applications.
- The available data do not provide formal Exposure at Default, institution-specific Loss Given Default, recovery rates, collection costs or profit margins.
- Financial-value estimates are scenario-based and should not be interpreted as realised or guaranteed savings.
- Observed relationships are predictive associations and should not automatically be interpreted as causal relationships.
- Employment category and other contextual variables require fairness review before operational use.
- Model performance may change as the economy, lending policy, borrower behaviour or data collection processes evolve.
- The test results support controlled piloting, not unrestricted deployment across all banking products and customer segments.

C.2 Minimum Monitoring Dashboard

Monitoring area	Indicator	Purpose
Predictive performance	PR-AUC, ROC-AUC, recall, precision, specificity	Detect deterioration in discrimination and default capture
Probability quality	Brier Score and calibration by risk band	Ensure probabilities remain meaningful
Error burden	False positives and false negatives	Measure customer and loss trade-offs
Portfolio stability	Share of applicants by risk band	Detect population or policy shifts
Business outcomes	Actual losses, recoveries, review costs, profitability	Validate real financial value

Operational efficiency	Review queue size, turnaround time, staff hours	Measure productivity benefit
Human oversight	Override frequency and reasons	Detect misuse or policy/model disagreement
Fairness	Borrower-group outcomes and adverse-action patterns	Detect inappropriate disparity
Drift	Changes in input distributions and realised default rates	Trigger investigation or recalibration

END OF FINAL REPORT